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SEPTEMBER 14, 1948
SUPERSEDING
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October 14, 1930

FEDERAL SPECIFICATION

HAIR; CURLED

This specification was approved by the Director, Bureau of Federal Supply for the use of all departments and establishments of the Government.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification is intended primarily to cover the requirements for hair to be used in the manufacture of furniture and mattresses.

1.2 Grades.—Curled hair covered by this specification shall be of the following grades, as specified:

Grade I.—Horse-tail hair.
Grade II.—Cattle-tail hair.
Grade III.—75 percent cattle-tail hair and 25 percent winter hog hair.
Grade IV.—50 percent cattle-tail hair and 50 percent winter hog hair.
Grade V.—30 percent cattle-tail hair and 70 percent winter hog hair.
Grade VI.—Horse-mane hair.

2. APPLICABLE SPECIFICATIONS

2.1 Specifications. — There are no other specifications applicable to this specification except those referenced in section 7 by individual departments.

3. REQUIREMENTS

3.1 Material. — The hair shall be well-curled and carded (or picker picked) and shall be resilient new stock. Horse-mane hair and horse-tail hair shall be clipped.

3.2 Odor.—The hair shall have no objectionable animal odor under any climatic condition, and shall conform to the requirements of 4.3.2.

3.3 Curl set and sterilization.—To insure proper set of curl and complete sterilization, the hair in the form of ropes shall be thoroughly steamed with 20 pounds of saturated steam at approximately 259° F. for not less than 20 minutes, or actively boiled in water for not less than 1 hour in such a manner as to insure uniform distribution of temperature. It shall be subsequently stored for not less than 15 days before carding, in such manner as to insure uniform drying.

3.3.1 In the process described in 3.3, prime winter hog hair shall be curled so as to conform in shape to the letters "c", "j", "u", or "o", or shall have a corkscrew twist having at least a half-circle form within the spiral loop.

3.4 Color.—The hair shall be of natural color and free from dye.

3.5 Length.

3.5.1 *Grade I.*—The length of the curled and carded grade I horse-tail hair shall be in accordance with the following table:

Not more than 10 percent by weight under 2 inches in length.
Not more than 60 percent by weight under 5 inches in length.
Not less than 40 percent by weight over 5 inches in length.

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3.5.2 *Grade II.*—The length of the curled and carded grade II cattle-tail hair shall be in accordance with the following table:

Not more than 12 percent by weight under 2 inches in length.

Not more than 50 percent by weight under 6 inches in length.

Not more than 90 percent by weight under 10 inches in length.

Not less than 10 percent by weight over 10 inches in length.

3.5.3 *Grade III.*—The length of the curled and carded grade III mixture of cattle-tail and winter hog hair shall be in accordance with the following table:

Not more than 32½ percent by weight under 2 inches in length.

Not more than 62 percent by weight under 6 inches in length.

Not more than 93 percent by weight under 10 inches in length.

Not less than 7 percent by weight over 10 inches in length.

3.5.4 *Grade IV.*—The length of the curled and carded grade IV mixture of cattle-tail and winter hog hair shall be in accordance with the following table:

Not more than 55 percent by weight under 2 inches in length.

Not more than 74 percent by weight under 6 inches in length.

Not less than 7 percent by weight over 9 inches in length.

3.5.5 *Grade V.*—The length of the curled and carded grade V mixture of cattle-tail and winter hog hair shall be in accordance with the following table:

Not more than 73 percent by weight under 2 inches in length.

Not more than 90 percent by weight under 5 inches in length.

Not less than 10 percent by weight over 5 inches in length.

3.5.6 *Grade VI.*—The length of the curled and carded grade VI horse-mane hair shall be in accordance with the following table:

Not more than 17 percent by weight under 2 inches in length.

Not more than 67 percent by weight under 5 inches in length.

Not less than 33 percent by weight over 5 inches in length.

3.6 Workmanship. — The hair shall be thoroughly washed, all epidermis removed, and permanently deodorized before spinning. There shall not be present any vegetable fibers, fetlock hair, hair-cloth clipping, combings, screenings, reworked hair, or hair from animals other than specified by the grade description. The hair shall be free from adherent scaly matter, dust, and foreign matters of every kind.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—Hair offered for delivery shall be divided at the place of manufacture into lots of 500 pounds and a portion of approximately 1 pound taken from each lot or fraction thereof. If the shipment exceeds 5000 pounds, it shall be divided into 10 lots of equal size and a portion of 1 pound taken from each such lot. All the portions thus taken shall be united into 1 sample, properly mixed, and forwarded to the points designated in the invitation for bids.

4.2 Inspection. — The hair shall be inspected with respect to curling, carding, cleanliness, and length.

4.2.1 *Length.* — The length of the hair shall be determined as follows: The sample shall be carefully opened and two 5-gram specimens taken for test. The hair comprising the specimen shall be picked out individually and measured along a ruler. Those 2 inches long and under shall be placed in a beaker marked "2 inches and under," and longer hairs placed in their respective beakers marked "2 inches to 5 inches," and "over 5 inches." Upon completion of this picking, the contents of each beaker shall be weighed and the percentage of each length based on

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the total weight of the specimen shall be calculated. The average of the two specimens shall represent the length of the hair in the delivery.

4.3 Tests.

4.3.1 *Identity.*—The identity of the hair shall be determined by the use of the microscope. Horse-tail hair may be distinguished from horse-mane hair by measuring the average diameter of selected samples of tail-hair; this average shall be not less than 0.0055 inches.

4.3.2 *Odor.*—Approximately 4 ounces of hair shall be immersed in 2 quarts of sterile distilled water, and shall remain for a period of not less than 2 hours after which it shall be thoroughly agitated by stirring, and then passed over a 14-mesh screen. Immediately upon removal of the hair, the water shall be clear and free from sediment. The water shall not develop an offensive odor when allowed to stand in an unsealed container for 71 to 73 hours at room temperature, 70° to 80° F.

4.3.3 *Dye.*—The presence of dye in the hair shall be determined by gently boiling a 5-gram specimen of the hair in 250 cubic centimeters of a 20-percent (by volume) solution of hydrochloric acid in a 400-cubic-centimeter beaker. If dyed hair is present a pink coloration will appear in a few minutes, the density of the coloration depending upon the amount of dyed hair present. A 25-gram specimen of hair shall also be boiled in a 2-percent solution of sodium carbonate or a 3 percent solution of ammonium hydroxide for one-half hour. If dyed hair is present, the solution will be darker than if only natural hair were present. When the presence of dyed hair is detected by these methods, some of the hairs shall be cross sectioned by means of a freezing microtome and examined microscopically to determine conclusively that the hair is dyed.

4.3.4 *Weight.*—At the time of weighing shipments at the place of delivery, specimens shall be taken from the interior of the bales, as described in 4.1, and the combined weight of the specimens or of a suitable portion of the specimens shall be taken. The combined specimens shall then be washed by stirring in a 1-percent solution of neutral soap in water at a temperature of 100° ± 5° F; after this wash, the sample shall be rinsed by stirring in water at a temperature of 100° ± 5° F, changing the water as necessary until it is not clouded or discolored after stirring. The sample shall then be brought to a constant dry weight at a temperature of 212° to 220° F and shall be weighed. The difference between the initial and final weights of the sample shall form the basis upon which the weight of the shipment, for purposes of payment, shall be calculated. (See 6.2.)

5. PREPARATION FOR DELIVERY

5.1 Packaging. — Unless otherwise specified, commercial packages are acceptable.

5.2 Packing.—Unless otherwise specified, each grade of hair shall be packed separately.

5.2.1 The subject commodity shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.3 Marking.—Unless otherwise specified, shipping containers shall be marked with the name of the item, the grade, and the net weight contained therein as defined by the contract or order under which shipment is made, the name of the manufacturer, the name of the contractor, and the number of the contract or order.

6. NOTES

6.1 Ordering information. — Purchasers should specify the grade required and should exercise any desired options offered herein. (See 1.2, 5.1, 5.2, and 5.3.)

6.2 Payment should be made on the basis of net weight delivered, unless the correction for dirt and moisture, determined as described in 4.3.4, exceeds 10 percent of the delivered weight, in which case twice the

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excess percentage should be deducted from the weight of the shipment for purposes of payment.

6.3 In order that the contractor may meet the requirements of 3.3 as to storing and carding and that there may be sufficient time for inspection, contracts for horsehair should not call for delivery before 30 days from date of contract or order, and purchasing officers should endeavor to anticipate their needs as far in advance as practicable and submit requests and requisitions accordingly.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following departmental specifications, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army.—U. S. Army Specification 100–2, Standard Specification for Marking Shipments by Contractors.

(Copies of specifications required for Army purchases may be obtained upon application to the procuring agency. Both the title and identifying symbol or number should be stipulated when requesting copies.)

7.3 Navy.

7.3.1 *Applicable specifications and other publications.*

Navy Department Specification:

General Specifications for Inspection of Material.

Bureau of Supplies and Accounts Publication:

Navy Shipment Marking Handbook.

(Copies of Federal specifications required for Navy purchases, Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained without cost upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. Both the title and identifying symbol or number should be stipulated when requesting copies of specifications.)

7.3.2 *Marking.*—In addition to any special marking required by the contract or order, marking of interior packages and shipping containers shall be in accordance with the requirements of the Navy Shipment Marking Handbook.

Navy: ShS

7.4 Air Force.—U. S. Army Specification 94–40645, Marking; Exterior, Domestic and Export Shipment, by Contractors.

(Copies of specifications required for Air Force purchases may be obtained from the Commanding General, Air Materiel Command, Wright-Patterson Air Force Base, Dayton, Ohio.)

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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